Sudarshan Shyam

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Education

Aarhus University — Aarhus, Denmark
PhD. in Computer Science — *Jan. 2023 – Jan. 2026*

University of California - San Diego — La Jolla, CA
Master of Science in Computer Science — *Sept. 2017 – June. 2019*

Indian Institute of Technology, Kharagpur — Kharagpur, India
Bachelor of Technology in Electrical Engineering — *Aug. 2013 – July. 2017*
Course Work: Machine Learning, Estimation of Signals & Systems, Computer Software

Work Experience

King's College, London — Incoming Postdoctoral Scholar
Intersection of AI Alignment and mechanism design — *September 2026-*

Microsoft Corporation — Software Development Engineer II
Bing & Substrate | Email Search | Index for Machine Learning based Search — *Dec 2020 - Dec 2021*

- Worked on Semantic search framework to power natural language user search queries for emails using Facebook AI Similarity Search (Faiss) algorithm.
- Improved Sentence Parsing mechanism using Blingfire to enable accurate and efficient extraction of sentences from an email for semantic search.
- Built tools for debugging the ingestion process to extract words and sentences extracted from an email intended for solving production issues.

Amazon — Software Development Engineer
eCommerce Foundational Technologies | Database Replication | Distributed and Parallel Systems — *Aug 2019 - Aug 2020*

- Redesigned the Record Versioning Algorithm for replication with a novel scheme of deduplication; enhancing the replication throughput by $\sim 96\%$. Design implementation was based on the concepts from concurrency and parallel computing.
- Spearheaded the activity on the evaluation of stability and robustness of the replication system using Gremlin- a Chaos Monkey like Amazon Resiliency Tool; sussed out potential weaknesses using quantitative analysis of performance data.
- Implemented basic quantum computing algorithms on Amazon Braket - a fully managed quantum computing service in Amazon and executed the ML models on live quantum computing hardware.
- Collaborated with the Amazon Braket team to improve the functionality of the newly launched quantum computing service. Designed and executed quantum machine learning algorithms on the platform.
- Contributed to developing a cross-storage replication platform; created an application to transform a record to different schemas according to customer use-cases using Database Transformation Languages.

Publications

- **Finding fair allocations under budget constraints**
  with Siddharth Barman, Arindam Khan, and K.V.N. Sreenivas
  AAAI 2023
- **Guaranteeing Envy-Freeness under Generalized Assignment Constraints**
  with Siddharth Barman, Arindam Khan, and K.V.N. Sreenivas
  EC 2023
- **Low-Distortion Clustering with Ordinal and Limited Cardinal Information**
  with Jakob Burkhardt, Ioannis Caragiannis, Karl Fehrs, Matteo Russo, and Chris Schwiegelshohn
  AAAI 2024
- **A new lower bound for multi-color discrepancy with applications to fair division**
  with Ioannis Caragiannis and Kasper Green Larsen
  SAGT 2025
- **Simple and Optimal Sublinear Algorithms for Mean Estimation**
  with Beatrice Bertolotti, Matteo Russo, and Chris Schwiegelshohn
  NEURIPS 2025

- **Existence and computation of fair allocations under constraints**
  with Siddharth Barman and Ioannis Caragiannis
  AAMAS 2026

- **Approximation Preserving Coresets**
  with Milind Prabhu and Chris Schwiegelshohn
  ICML 2026

- **Coresets for Capacitated Clustering via Dual Concentration**
  with Milind Prabhu
  Under Submission

Ongoing work

Studying and controlling the behavior of LLMs in markets · Coresets for Clustering problems

Research Experience

University of Illinois Urbana-Champaign — Research Assistant
Game Theory | Fair Division | Computational Social Choice — *Aug 2024 - Dec 2024*
Advisors: Prof. Ruta Mehta

- Working on problems related to fair division of indivisible goods for the fairness notion called EFX.
- Studied the literature on core-stable committees and worked on finding simple algorithms for special cases.

Indian Institute of Science — Research Associate
Game Theory | Fair Division | Approximation Algorithms — *Jan 2022 - Dec 2022*
Advisors: Prof. Siddharth Barman, Prof.Arindam Khan

- Worked on problems related to fair division of goods among multiple agents under constraints. Designed efficient algorithms and proved correctness and fairness guarantees.
- Worked on optimization problems related to job scheduling/load balancing with respect to general convex norms.
- Studied optimization problems related to job scheduling and load balancing for the objectives of general convex norms.

University of California - San Diego — Graduate Student Researcher
Boosting | Complexity Theory — *Summer 2018*
Advisor: Prof. Russell Impagliazzo

- Analyzed and evaluated the conditions for applying multiplicative weights method style boosting.
- Worked on boosting algorithms based on GANs (Generative Adversarial Networks) and their applications.

University of California - San Diego — Graduate Student Researcher
Dimensionality Reduction — *Sept 2018 - Jan 2019*
Advisor: Prof. Yoav Freund

- Studied Dimensionality reduction techniques and the effect of noise on their performance.
- Worked on improving the noise sensitivity of modified alternating diffusion, a manifold learning technique.

National University of Singapore — Research Intern
Quantum Computing | Sum of Squares — *May 2016 - July 2016*
Advisor: Prof.Rahul Jain

- Studied the relationship between Hypercontractivity and Small Set Expansion Problem and how the Sum of Squares Hierarchy can be used to refute the Unique Games Conjecture.

Projects

Chess Engine

Personal Project | Algorithms | Deep Learning | Reinforcement Learning *Jan 2020 - Dec 2020*

- Developed a working chess engine using Minimax search algorithm using alpha-beta pruning to reduce complexity.
- Experimented with reinforcement learning techniques to simulate the recent success of AlphaZero and Leela Chess Zero, with limited computing resources.

Text recommendation engine

Personal Project | Algorithms | Recommender Systems | Tensorflow | Keras *Jan 2013 - July 2019*

- Designed and implemented Word Cue, a custom text recommendation engine for Android P2P messaging app. Advanced String algorithms - KMP and Wagner Fischer's, and machine learning models - Word2vec, CNN's were used to improve performance and recommendations.

Awards

- Ranked 9th among more than 10,000 teams in ACM-ICPC Regionals (International Algorithmic Problem Solving Contest).
- Actively participated and won Inter-College Math Olympiads and Algorithmic Programming competitions. Also gave lectures and tutorial sessions to to help juniors get introduced to competitive programming and math contests.

Teaching Experience (Teaching Assistant)

- Discrete Mathematics, Winter 2018
- Discrete Optimization Algorithms, Fall 2018
- Graduate Algorithms Fall 2018
- Theory of Computation, Winter 2019
- Computational Complexity Theory, Spring 2019

Relevant Courses

- Communication Complexity
- Quantum Computation
- Algorithmic Graph Theory
- Enumerative Combinatorics
- Randomized Algorithms
- Information Theory and Error Correcting Codes
- Advanced Complexity Theory

Competencies

Languages: C, C++, Java, Python, SQL, Ruby, OCaml

Libraries : TensorFlow, PyTorch, Numpy, Keras, Pandas